averageInteractionsPerCrossing

overview · all regions and selections · rank 57 / 169

rank 57 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.368	0.373	0.338	0.388	0.367	0.355	0.380	0.391	34,970
endcap	ht40	0.407	0.412	0.376	0.426	0.405	0.394	0.419	0.430	7,390
endcap	bkg split	0.051	below 27,580 / above 7,390							
barrel	all	0.326	0.324	0.307	0.367	0.315	0.324	0.316	0.339	1,089
barrel	ht40	0.396	0.400	0.369	0.453	0.397	0.399	0.393	0.411	220
barrel	bkg split	0.118	below 869 / above 220							
all	all	0.370	0.374	0.338	0.396	0.365	0.361	0.374	0.391	36,059
all	ht40	0.410	0.413	0.377	0.434	0.404	0.401	0.414	0.431	7,610
all	bkg split	0.052	below 28,449 / above 7,610							

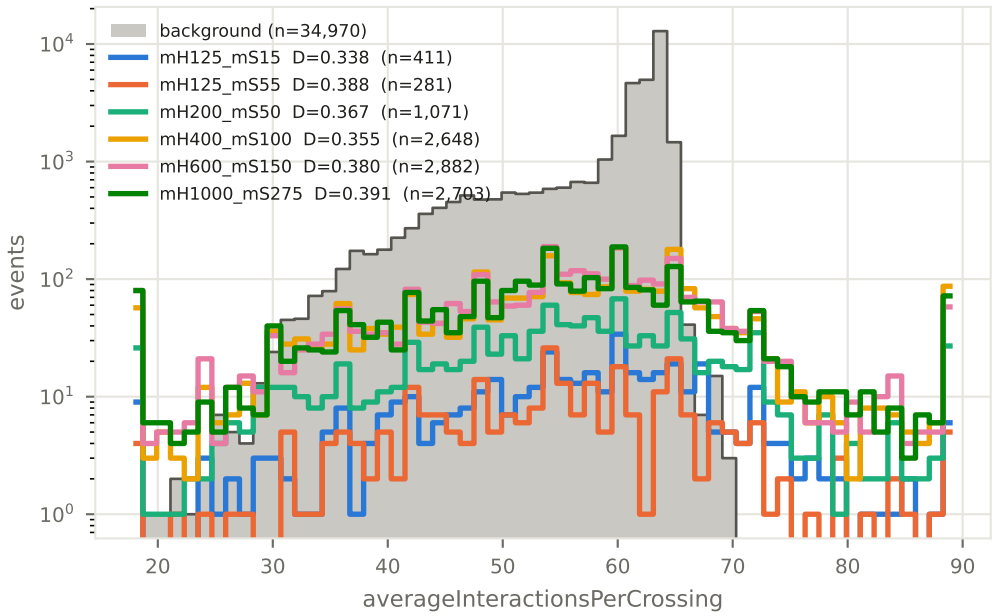
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

averageInteractionsPerCrossing

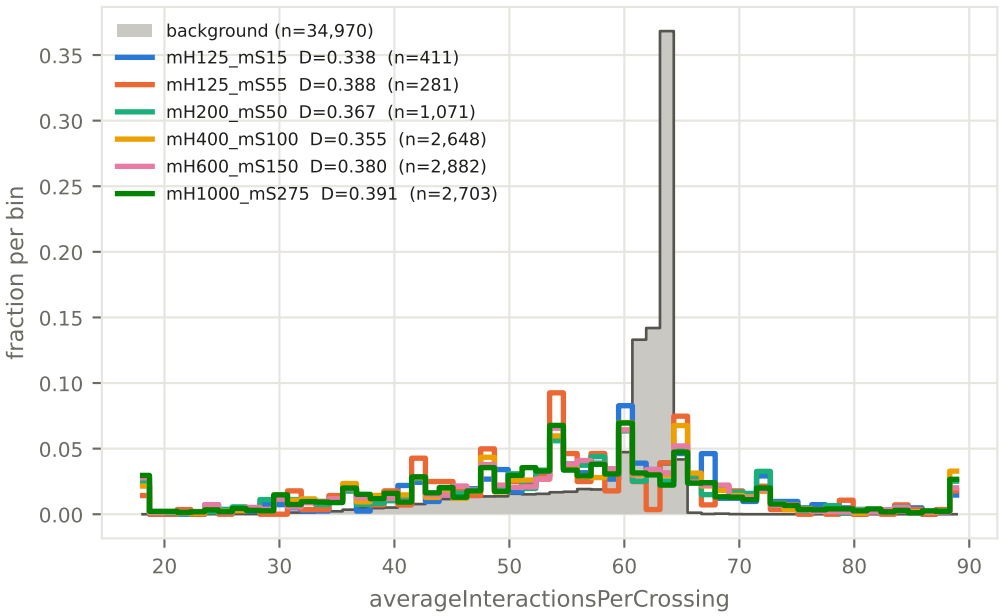
region: ENDCAP · selection: all (every event) · rank 57 / 169

D_equal 0.368 D_pooled 0.373 min 0.338 (mH125_mS15) max 0.391 (mH1000_mS275) bkg n = 34,970 sig n = 9,996

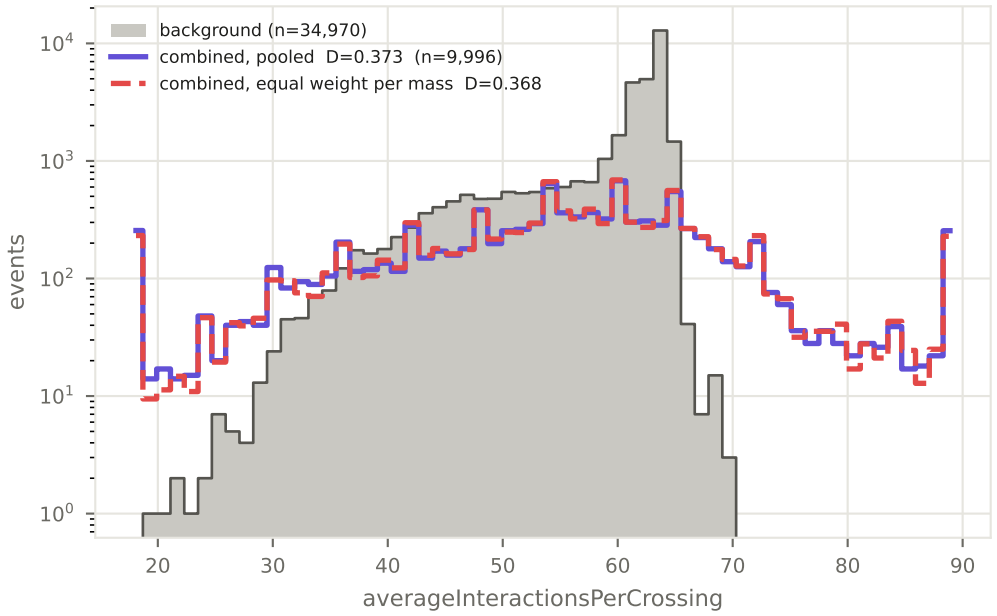
each mass point vs background · raw



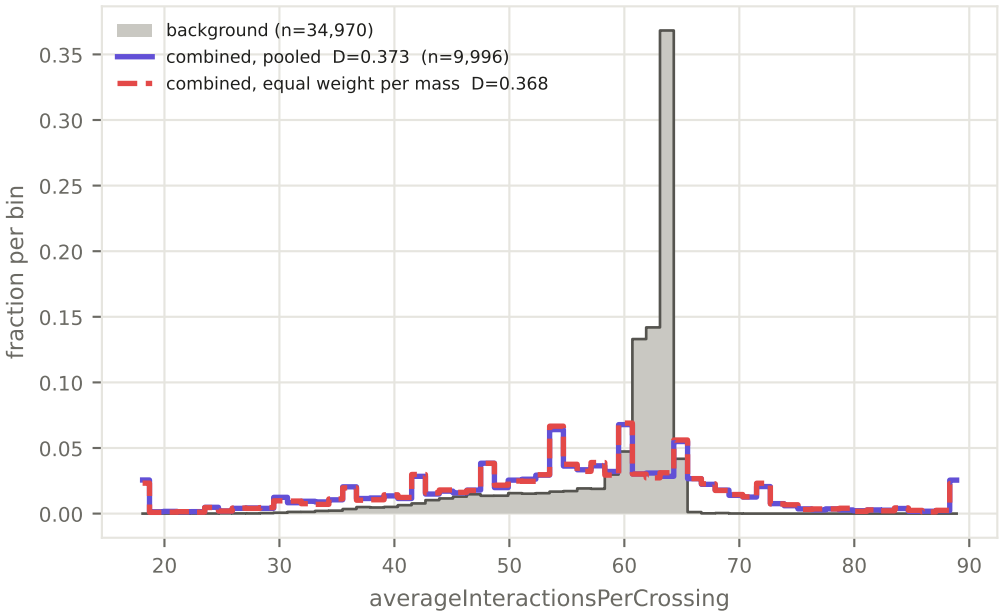
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



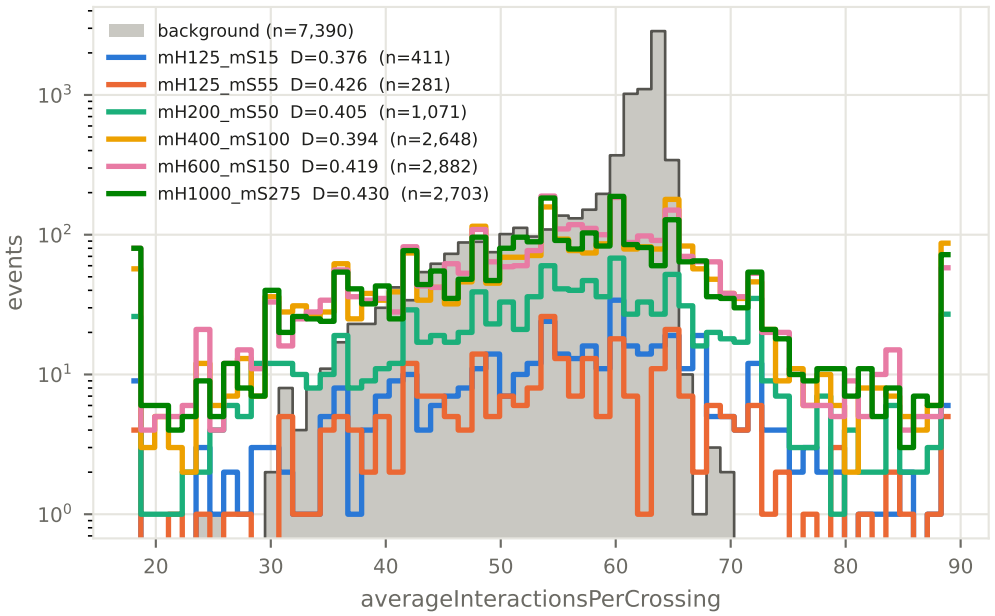
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

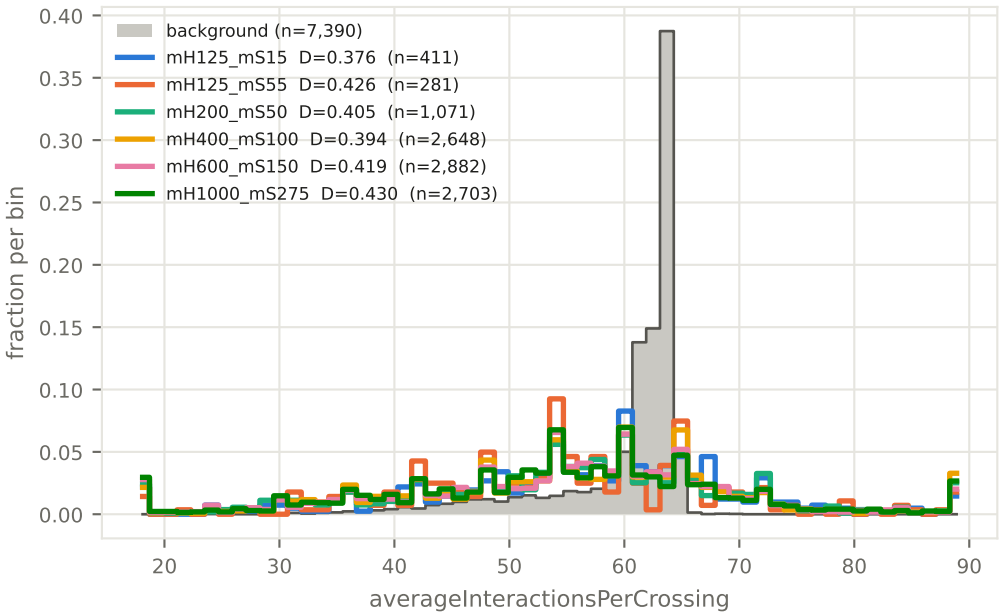
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D_equal 0.407 D_pooled 0.412 min 0.376 (mH125_mS15) max 0.430 (mH1000_mS275) bkg n = 7,390 sig n = 9,996

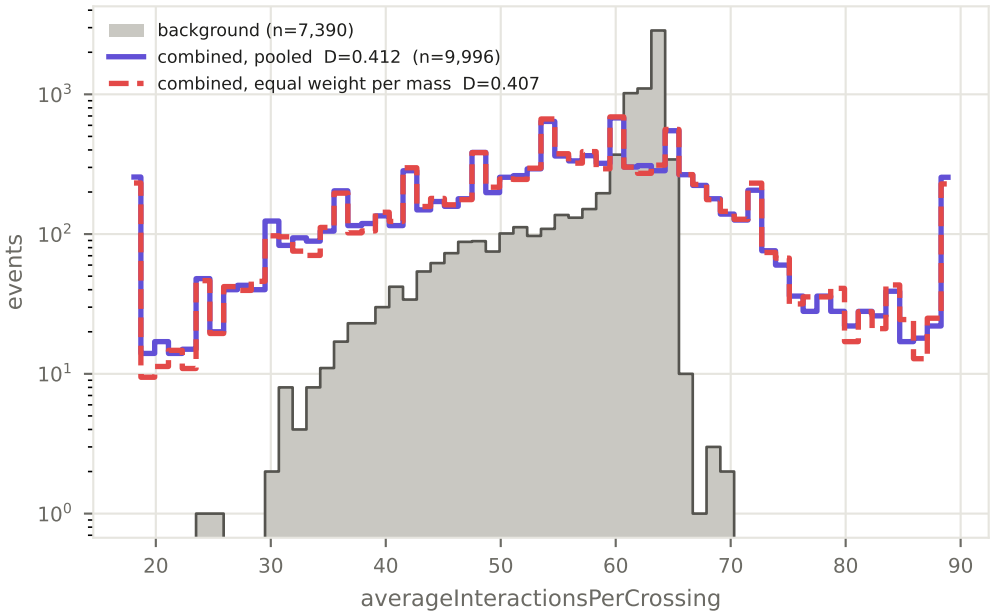
each mass point vs background · raw



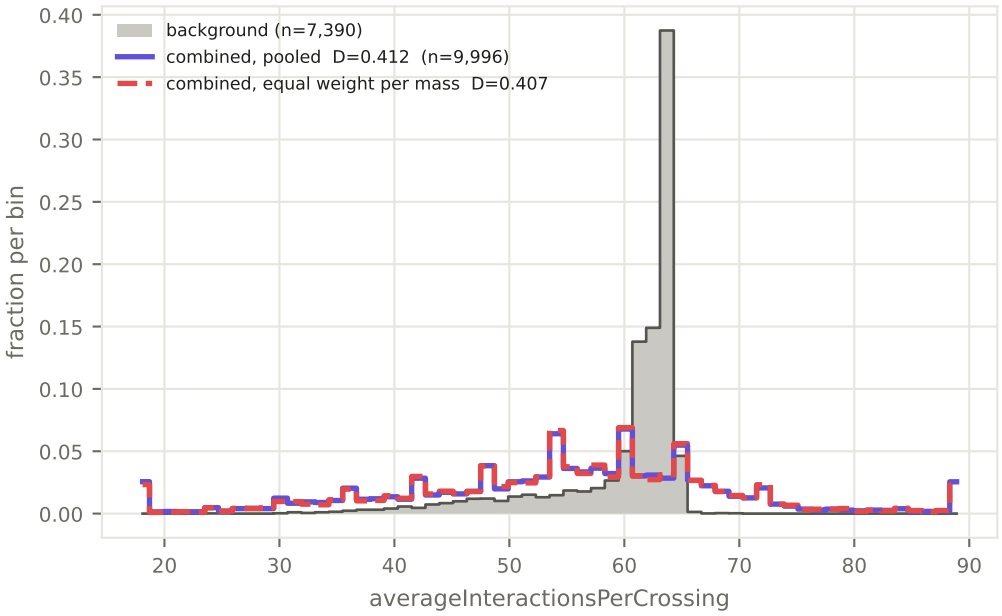
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



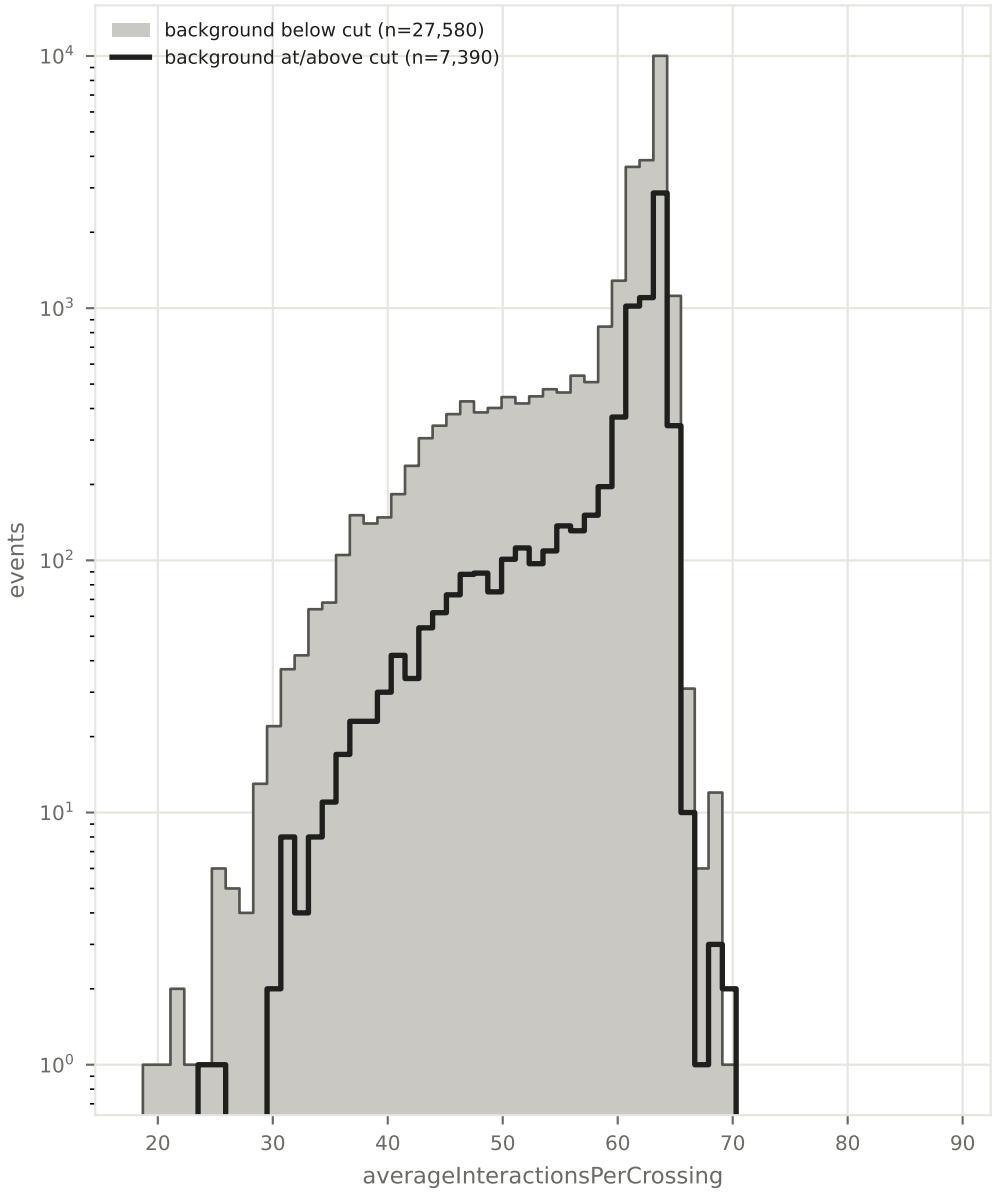
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

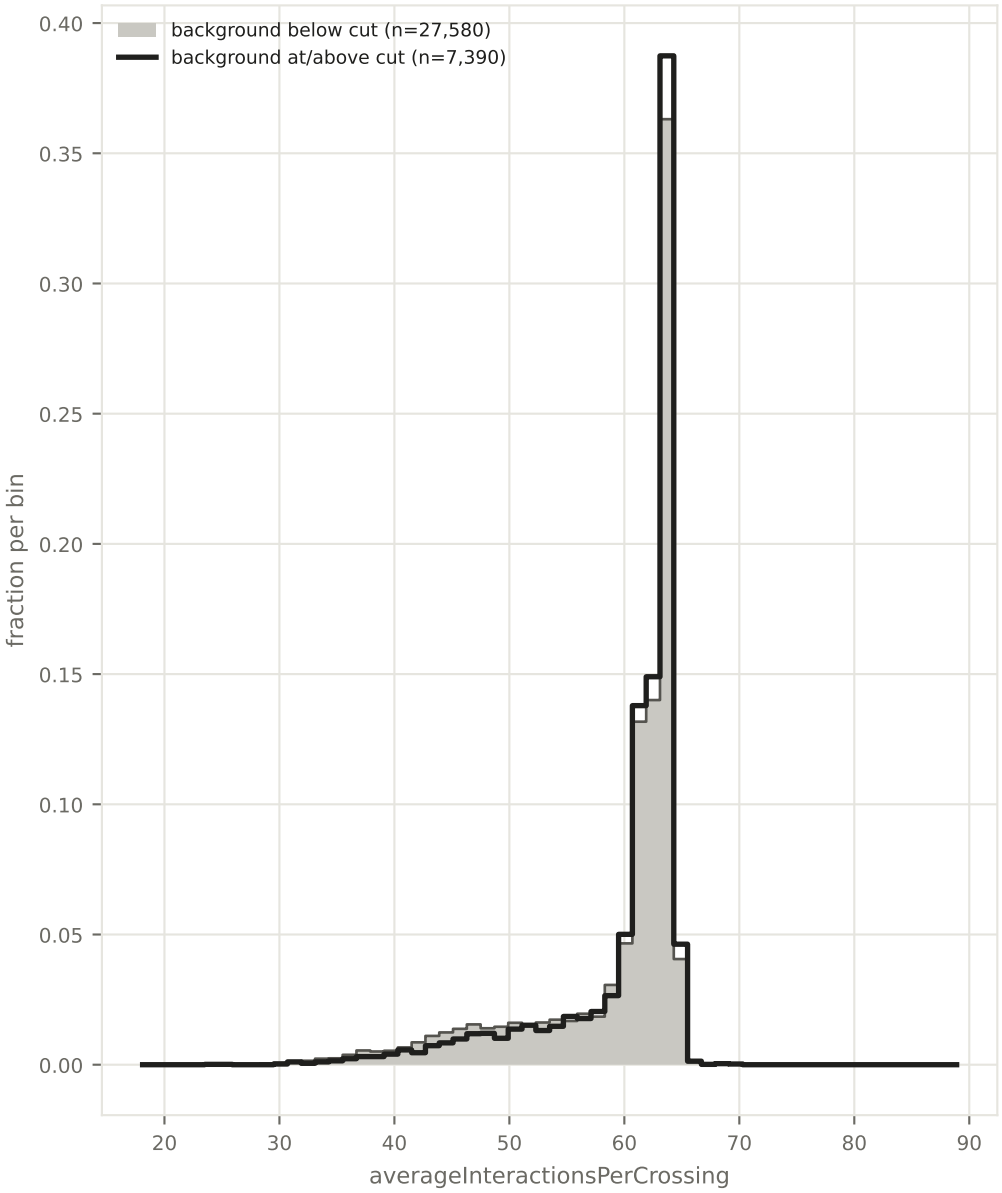
region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D(below vs above) 0.051 n below = 27,580 n above = 7,390 large D = background shape changes across the cut

background below vs at/above · raw



background below vs at/above · normalized

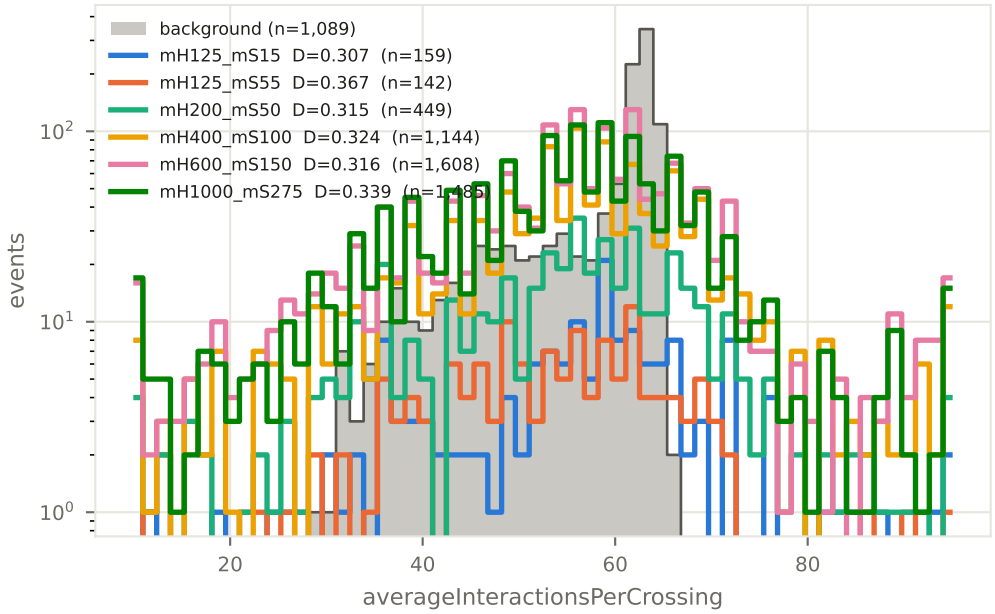


averageInteractionsPerCrossing

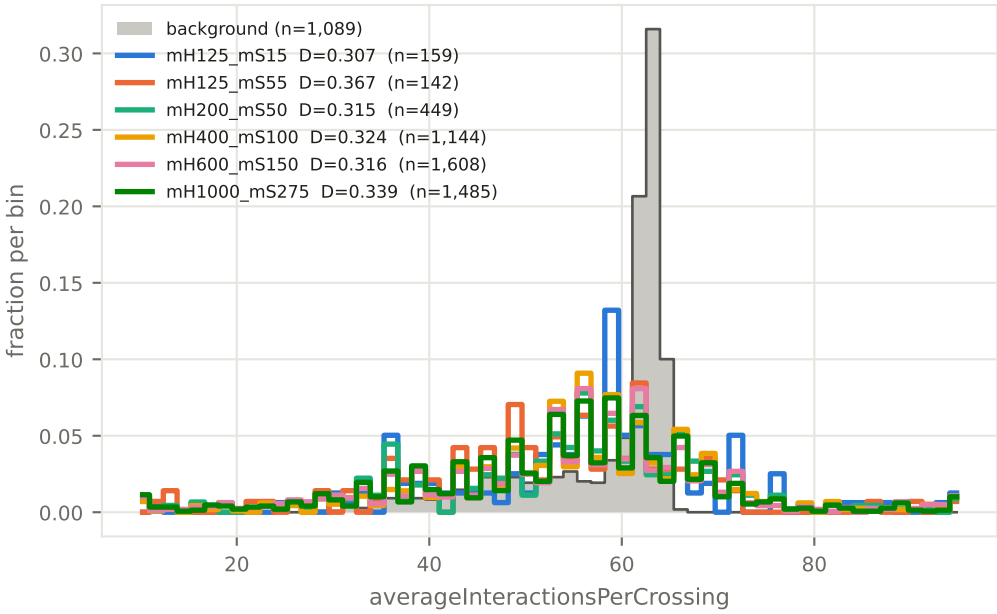
region: BARREL · selection: all (every event) · rank 57 / 169

D_equal 0.326 D_pooled 0.324 min 0.307 (mH125_mS15) max 0.367 (mH125_mS55) bkg n = 1,089 sig n = 4,987

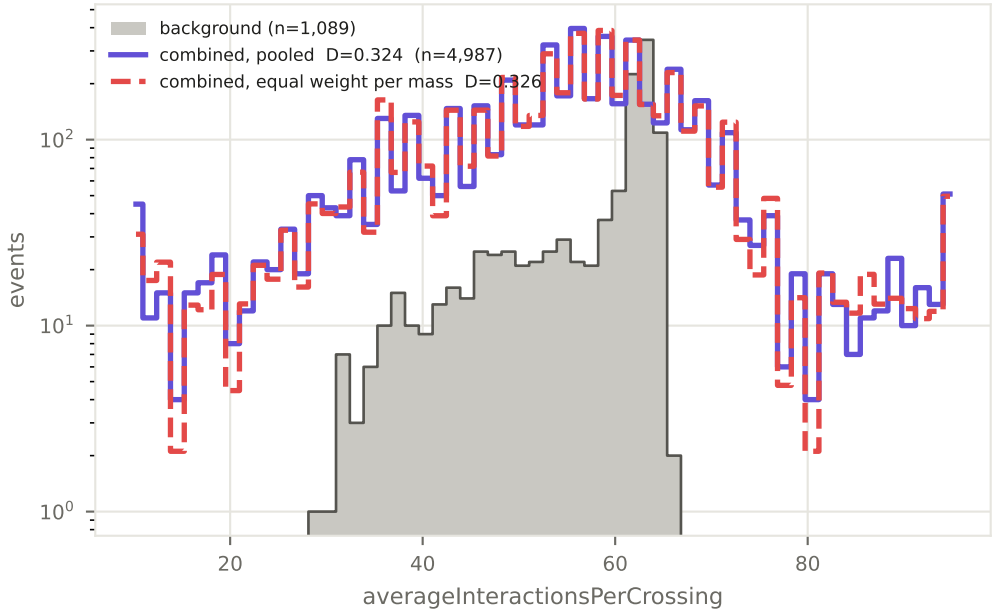
each mass point vs background · raw



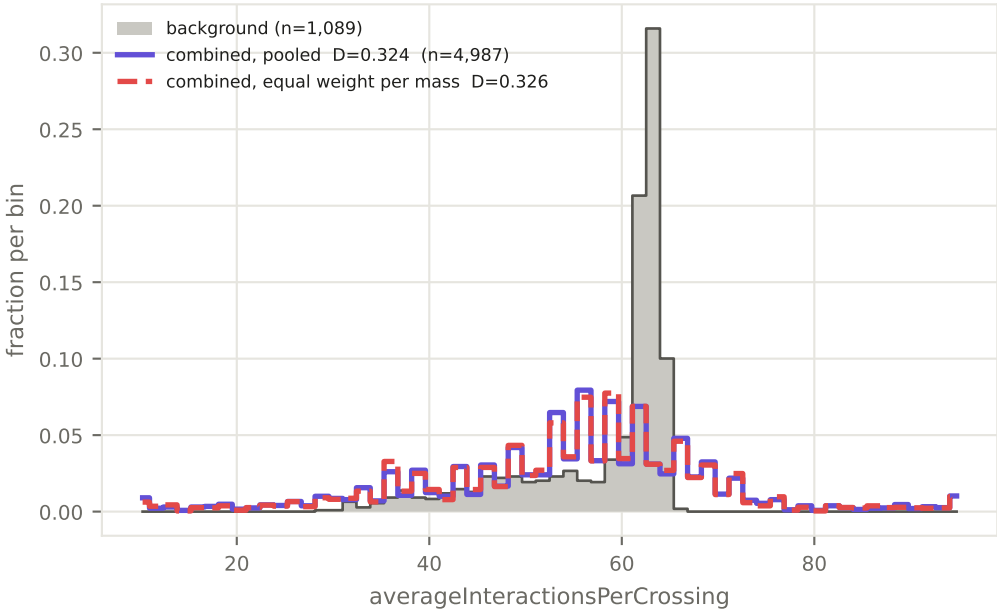
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



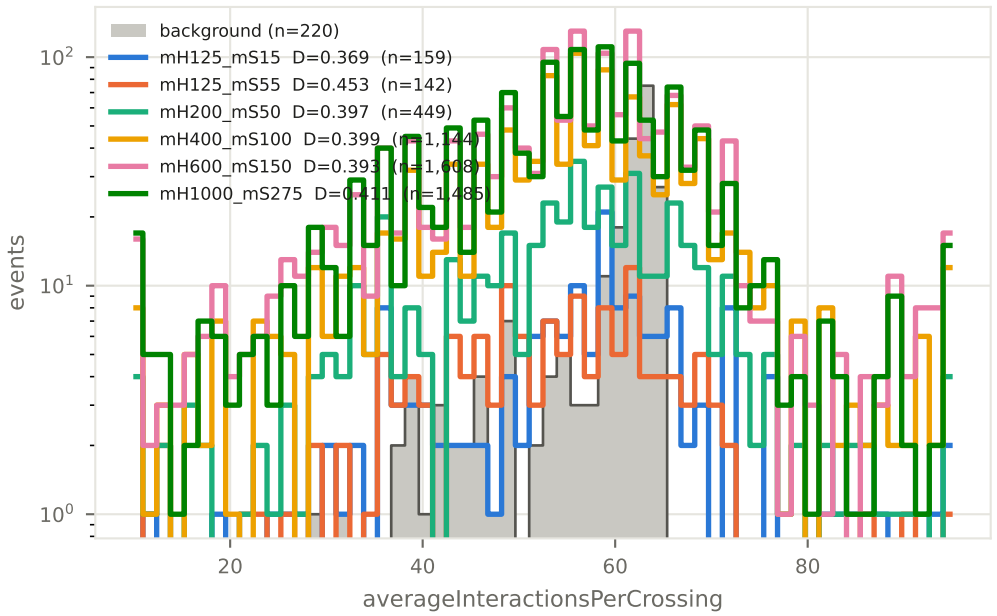
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

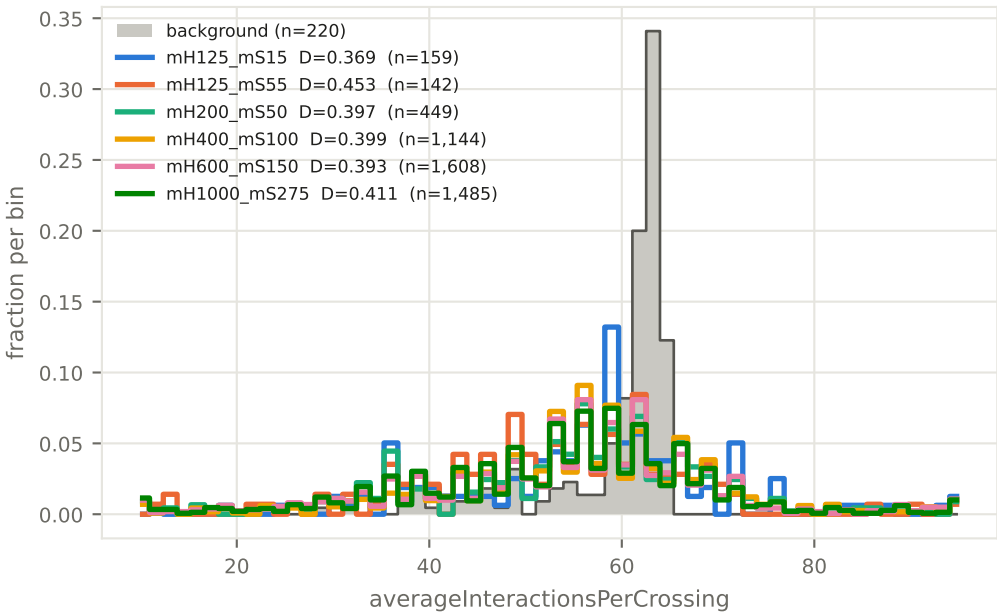
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D_equal 0.396 D_pooled 0.400 min 0.369 (mH125_mS15) max 0.453 (mH125_mS55) bkg n = 220 sig n = 4,987

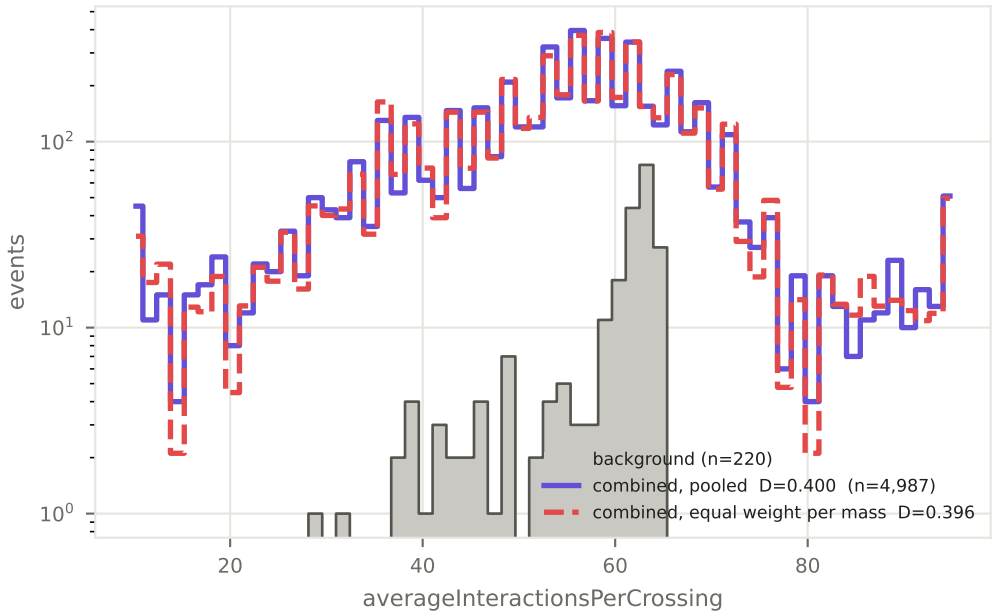
each mass point vs background · raw



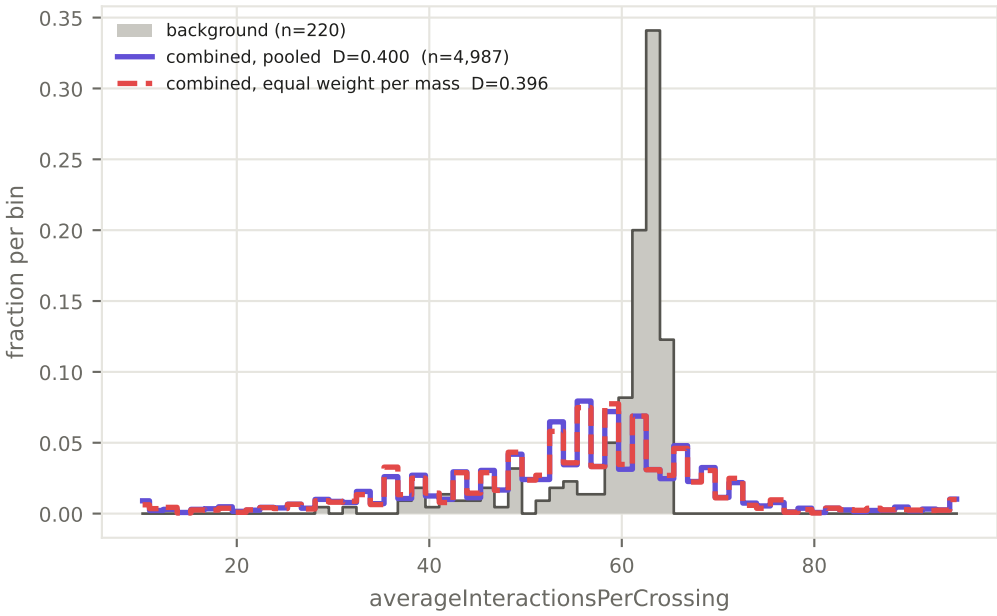
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

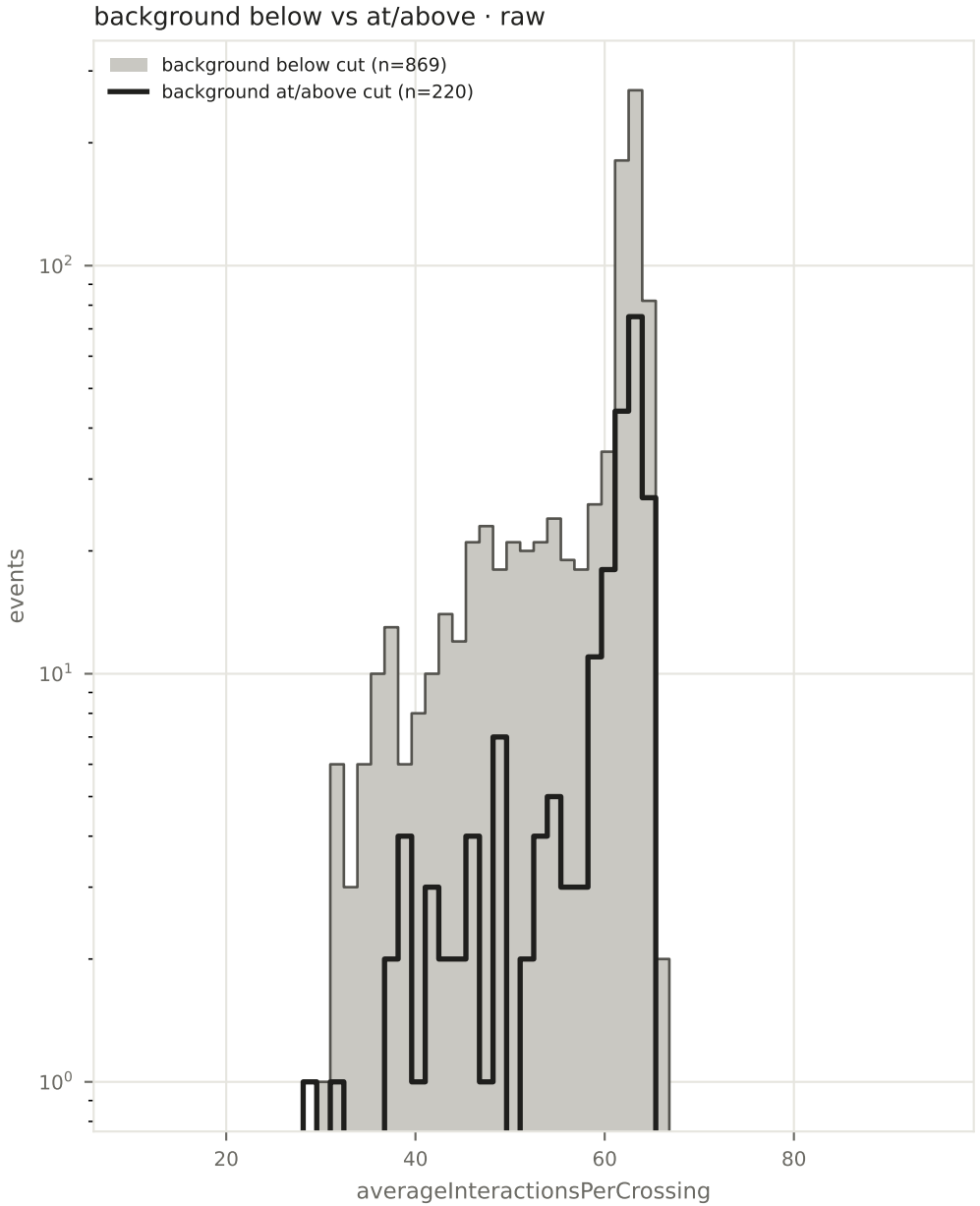


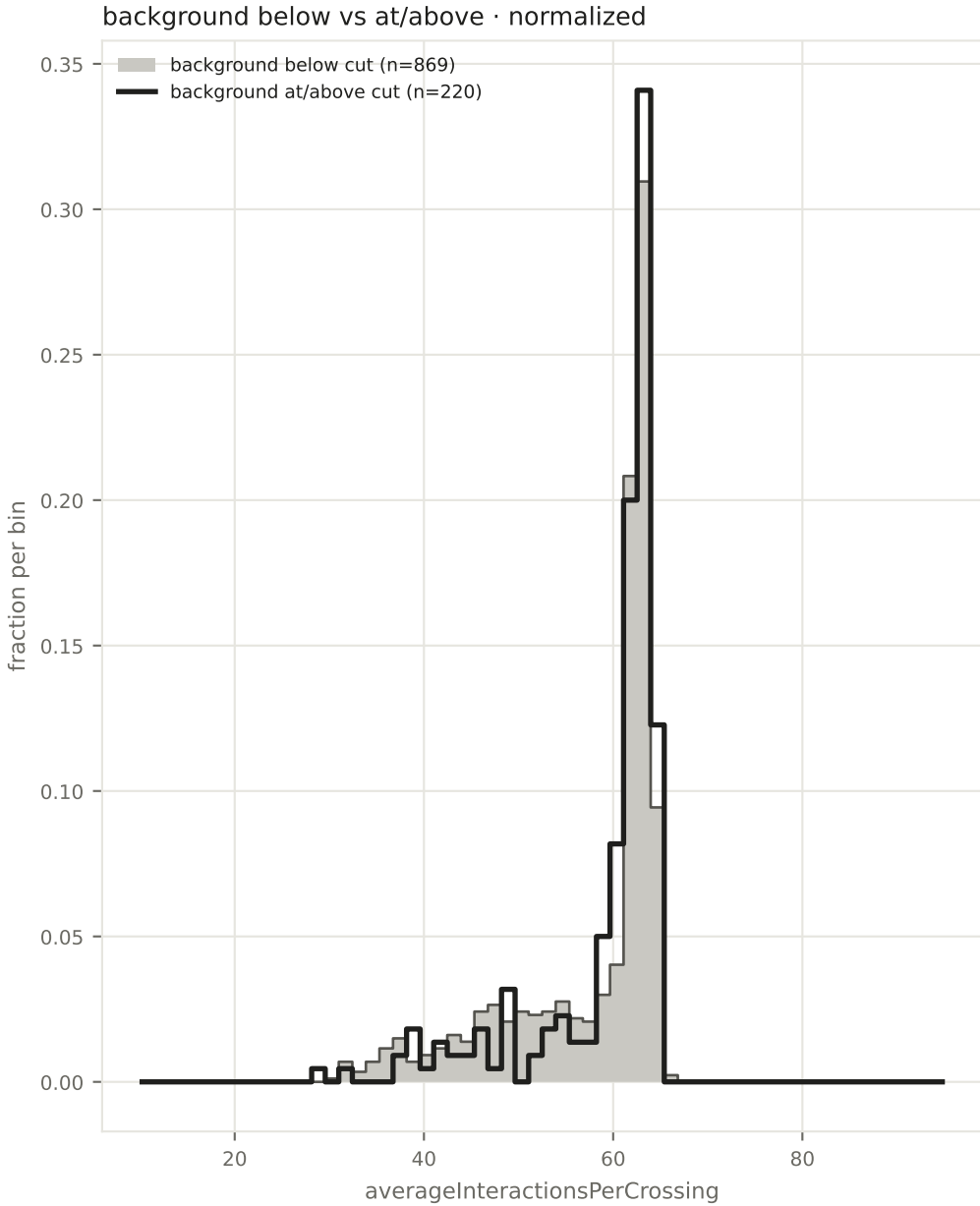
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D(below vs above) 0.118 n below = 869 n above = 220 large D = background shape changes across the cut



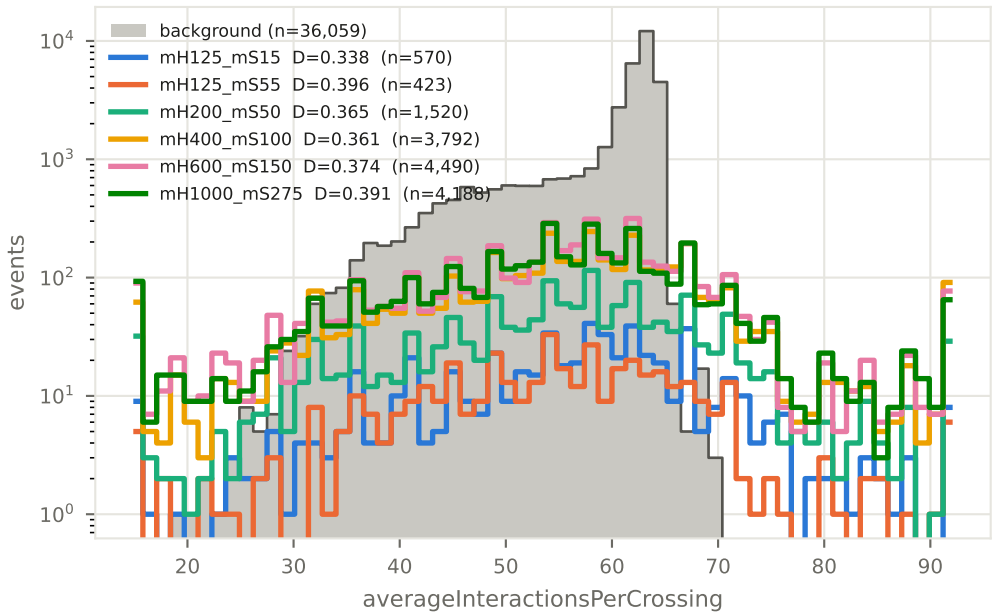


averageInteractionsPerCrossing

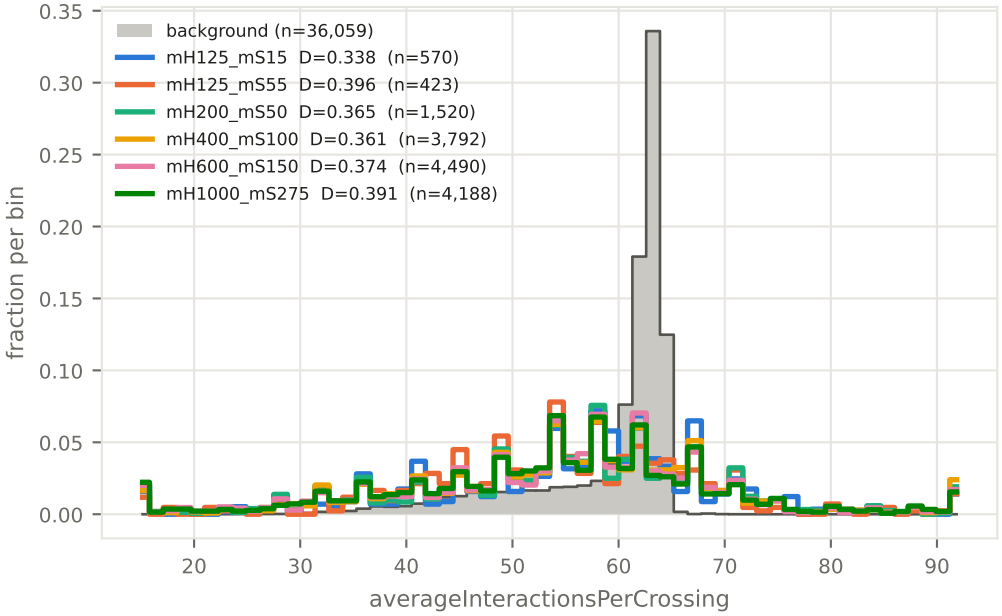
region: ALL · selection: all (every event) · rank 57 / 169

D_equal 0.370 D_pooled 0.374 min 0.338 (mH125_mS15) max 0.396 (mH125_mS55) bkg n = 36,059 sig n = 14,983

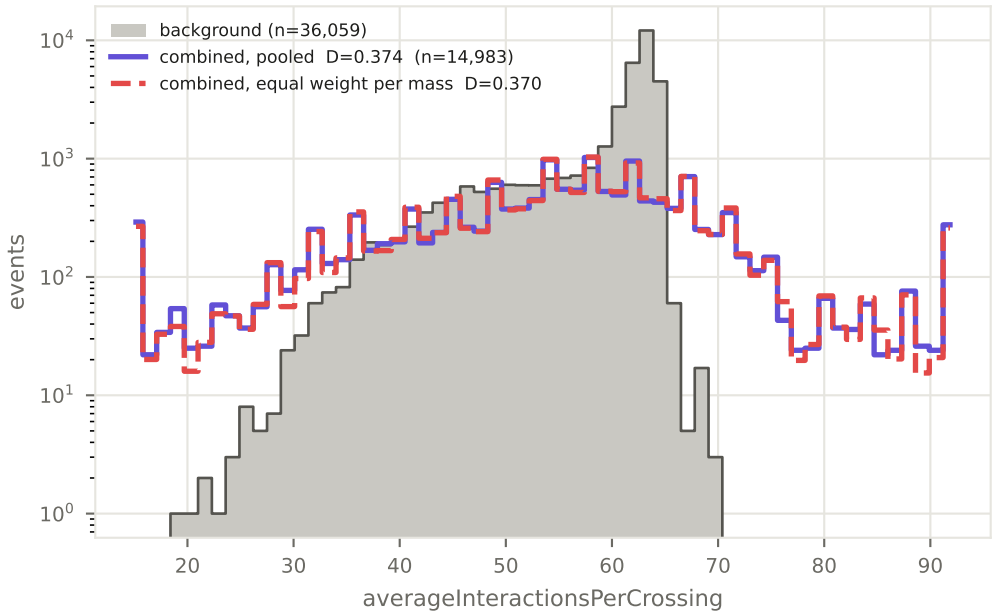
each mass point vs background · raw



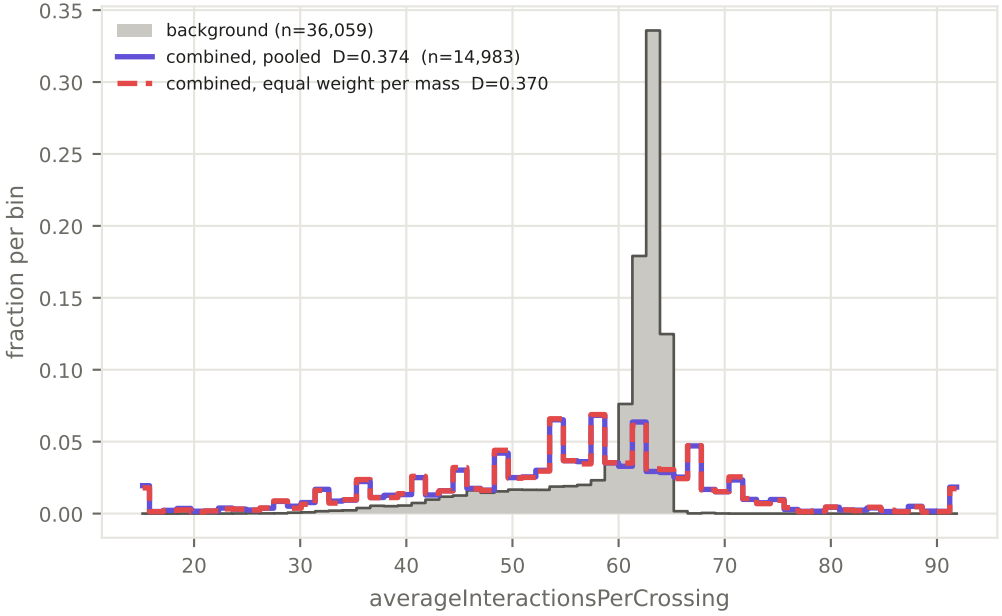
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



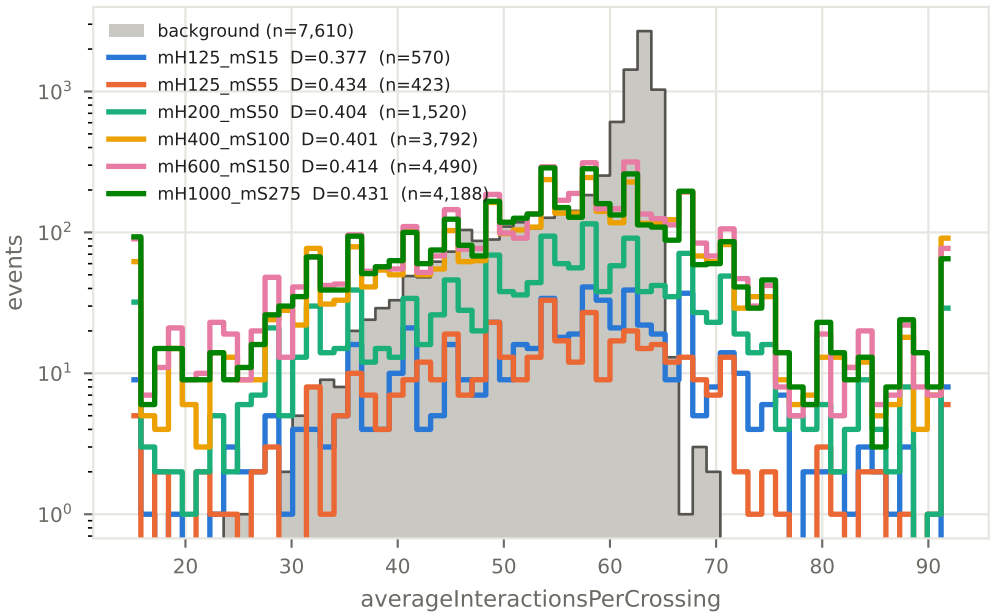
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

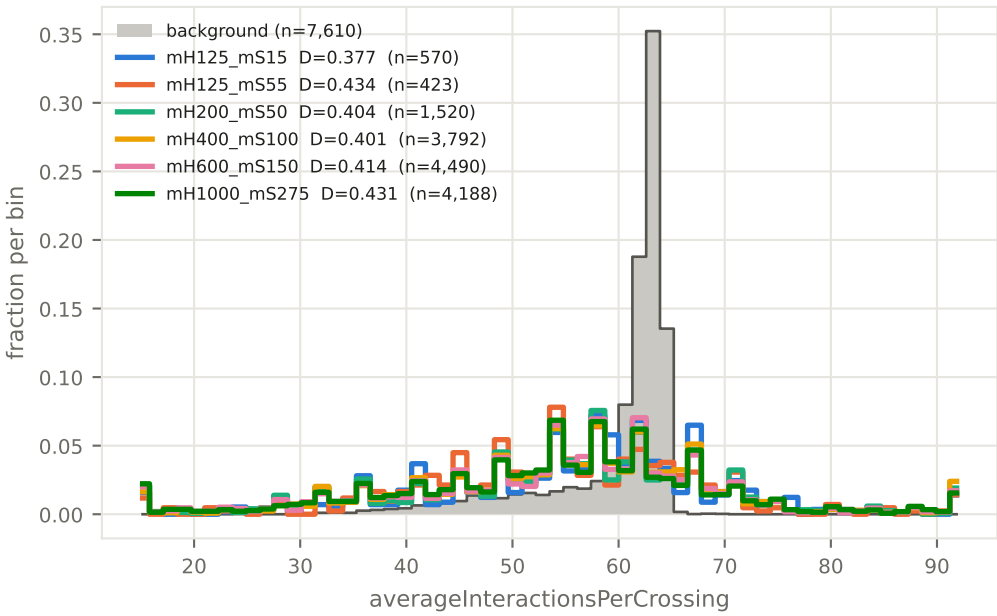
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D_equal 0.410 D_pooled 0.413 min 0.377 (mH125_mS15) max 0.434 (mH125_mS55) bkg n = 7,610 sig n = 14,983

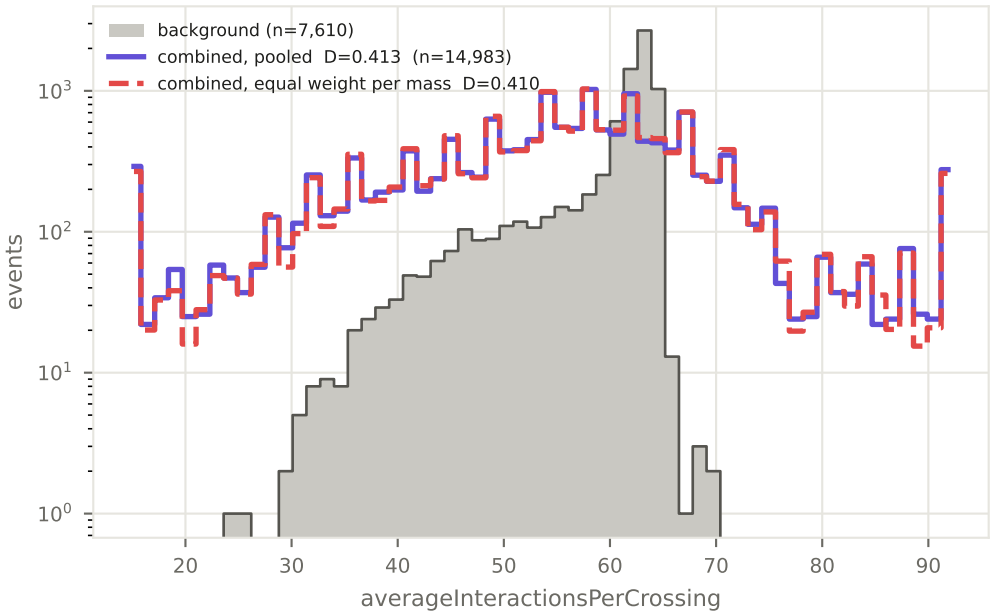
each mass point vs background · raw



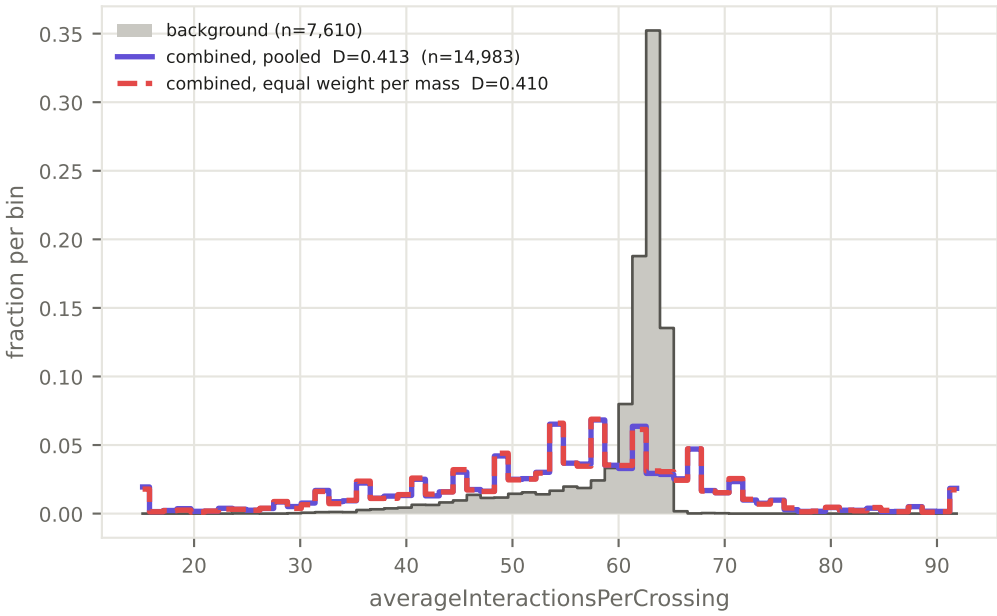
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



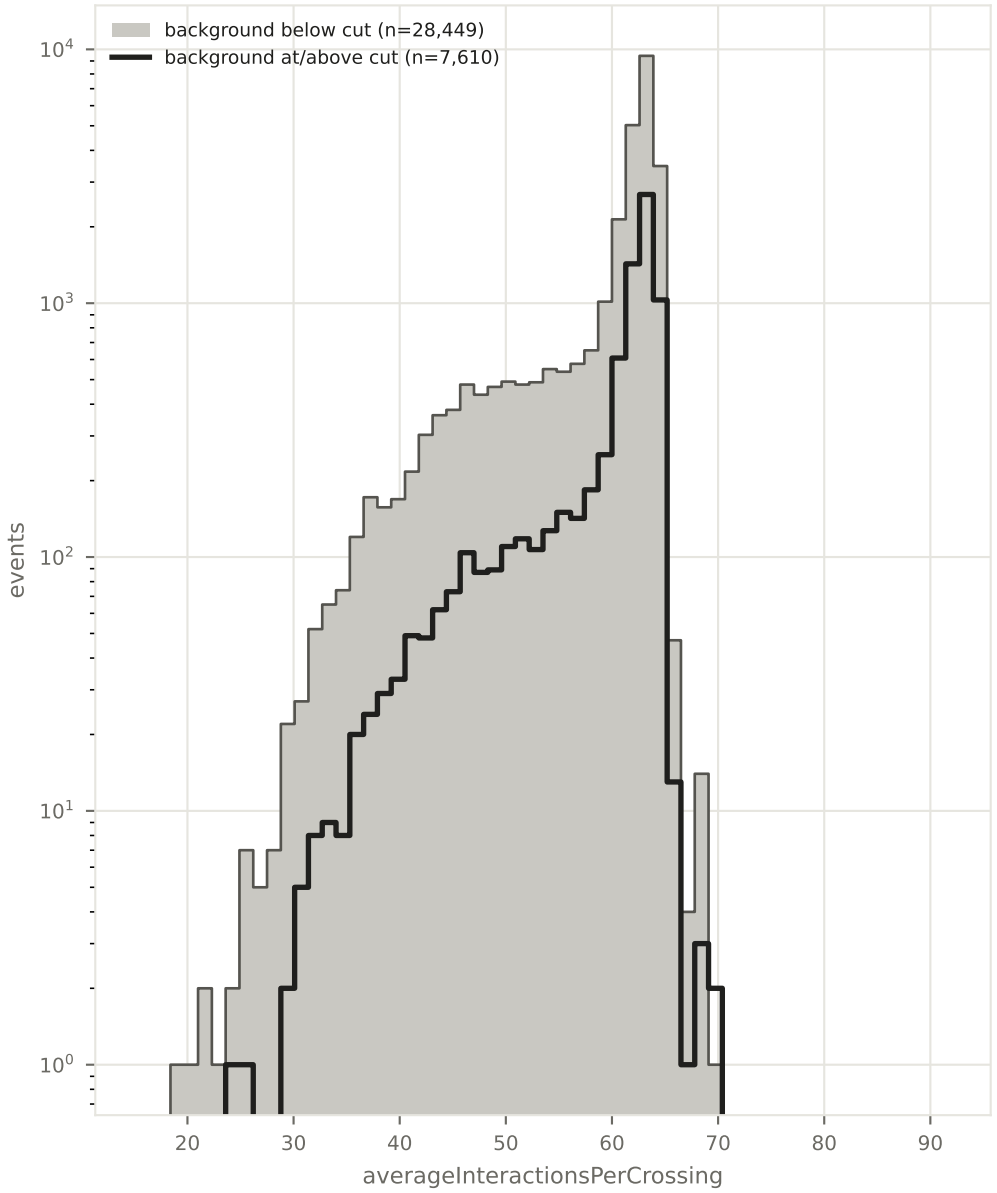
edge bins include under/overflow · NaN and -999 removed

averageInteractionsPerCrossing

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 57 / 169

D(below vs above) 0.052 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

background below vs at/above · raw



background below vs at/above · normalized

